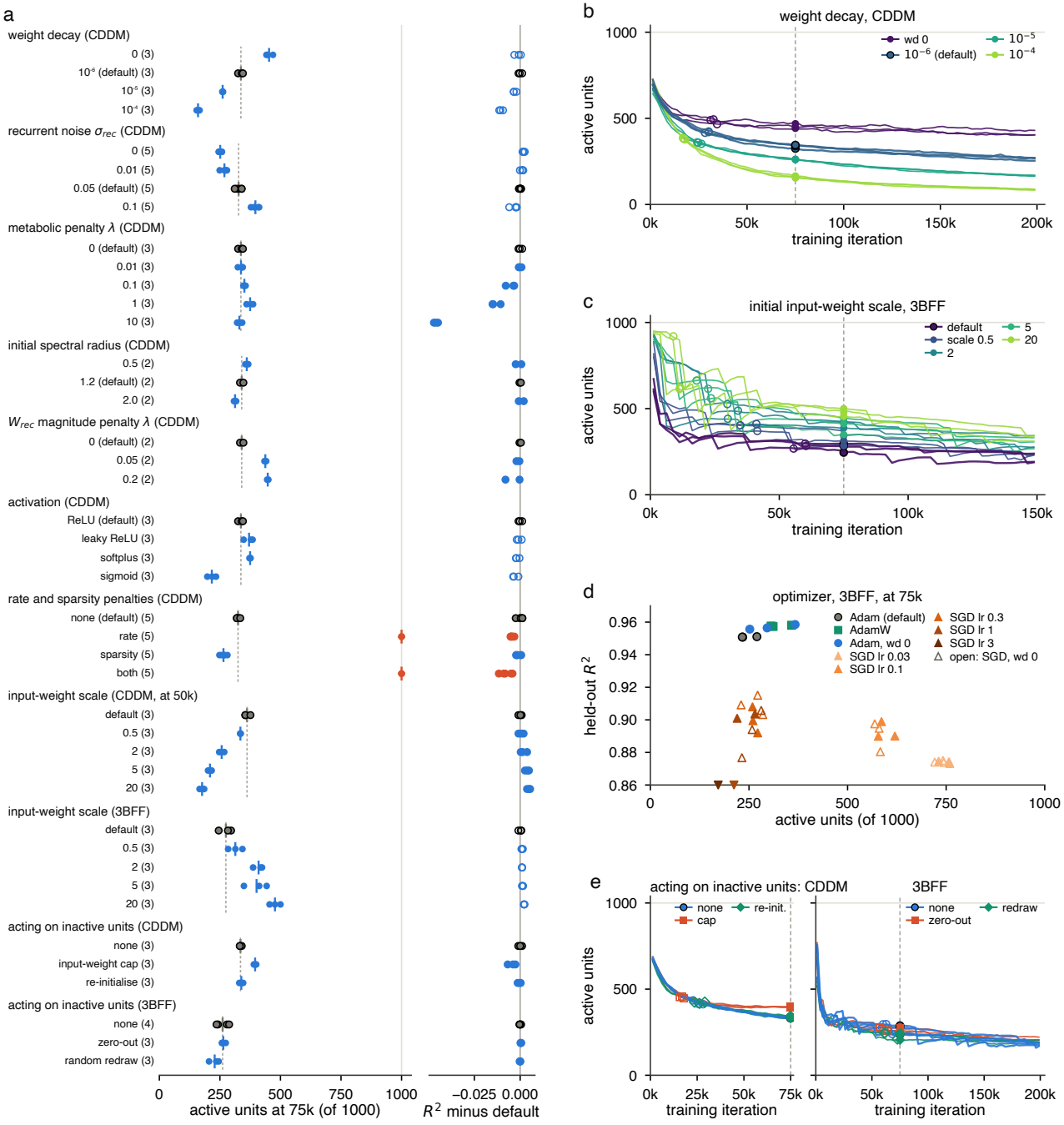

a
weight decay (CDDM)
0 (3)
10⁻⁶ (default) (3)
10⁻⁵ (3)
10⁻⁴ (3)
recurrent noise σrec (CDDM)
0 (5)
0.01 (5)
0.05 (default) (5)
0.1 (5)
metabolic penalty λ (CDDM)
0 (default) (3)
0.01 (3)
0.1 (3)
1 (3)
10 (3)
initial spectral radius (CDDM)
0.5 (2)
1.2 (default) (2)
2.0 (2)
Wrec magnitude penalty λ (CDDM)
0 (default) (2)
0.05 (2)
0.2 (2)
activation (CDDM)
ReLU (default) (3)
leaky ReLU (3)
softplus (3)
sigmoid (3)
rate and sparsity penalties (CDDM)
none (default) (5)
rate (5)
sparsity (5)
both (5)
input-weight scale (CDDM, at 50k)
default (3)
0.5 (3)
2 (3)
5 (3)
20 (3)
input-weight scale (3BFF)
default (3)
0.5 (3)
2 (3)
5 (3)
20 (3)
acting on inactive units (CDDM)
none (3)
input-weight cap (3)
re-initialise (3)
acting on inactive units (3BFF)
none (4)
zero-out (3)
random redraw (3)
0
250
500
750
1000
active units at 75k (of 1000)
−0.025
0.000
R² minus default
b
weight decay, CDDM
wd 0
10⁻⁶ (default)
10⁻⁵
10⁻⁴
active units
training iteration
c
initial input-weight scale, 3BFF
default
scale 0.5
2
5
20
d
optimizer, 3BFF, at 75k
Adam (default)
AdamW
Adam, wd 0
SGD lr 0.03
SGD lr 0.1
SGD lr 0.3
SGD lr 1
SGD lr 3
open: SGD, wd 0
held-out R²
active units (of 1000)
e
acting on inactive units: CDDM
3BFF
none
re-init.
cap
redraw
zero-out
active units
training iteration